

2026-02-24-25 Run SuperDec experiments on CLEVR-3D with cuboid loss

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24.2. Implement normals computation and start fist experiments

Using Claude Code, I compute normals from the GT depth maps in the CLEVR-3D dataloader. Based on these normals, I compute the cuboid loss.

How training is done

- again 1000 epochs (train dataset has 34963 samples, hence on a single H100 with a batch size 32, there are 1093 iterations per epoch and with 4 H100s each batch size 32, there are 274 iterations per epoch)
- evaluate every epoch (val dataset has 52 samples)
- save checkpoint every epoch

After each epoch, the average training loss and learning rate are logged.

After each epoch, `self.evaluate` is called and the validation output dict is logged.

Trainings are started with Imbrun with `train_cleivr3d.yaml` config. Experiments differ in terms of values in the config files and source code (use git diff on the tags to see differences).

How evaluation is done

Quantitative evaluations

Currently I didn't run quantitative evaluations for CLEVR-3D experiments yet.

Visualizations

All I did was look at visualizations:

For visualizations, I first do `rm -dr output_npz`

Then edit `save_npz_cleivr3d.yaml` (setting `checkpoints_folder` and `checkpoints_file`). Note: make sure that the code (not the `train_cleivr3d.yaml` config ; it does not matter as its anyways loaded from the checkpoint folder) fits to the experiment; some experiments had hacky things in the code!

Then run the vscode `Save npz CLEVR-3D launch config`

After that I run the `Visualize objects CLEVR-3D` launch config in vscode and open Viser in my browser.

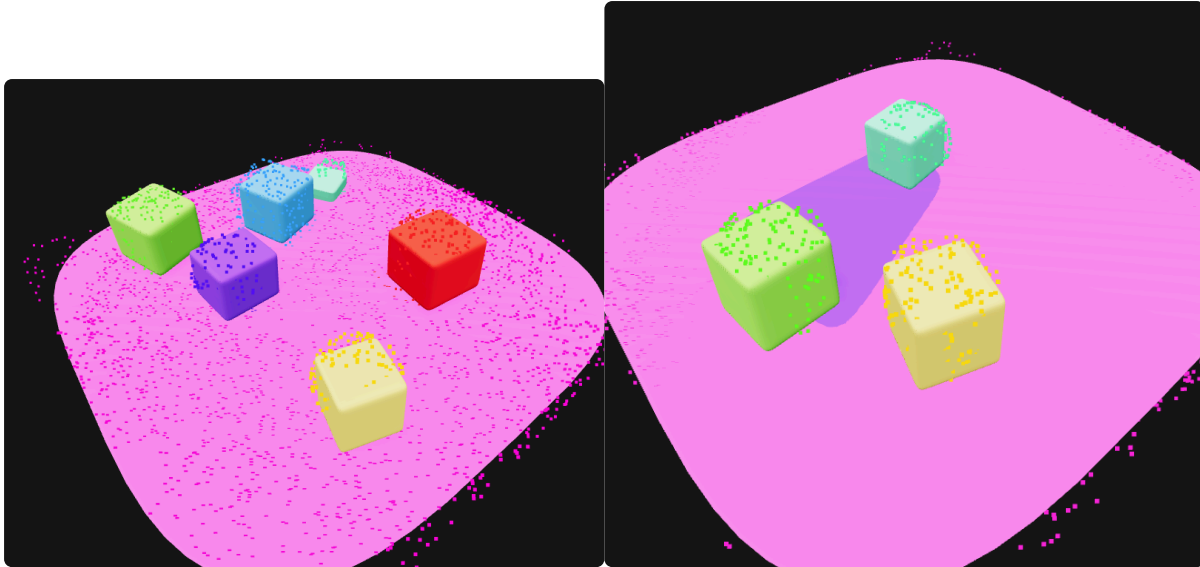
Experiments:

- `superdec_cleivr3d_000008` : chamfer loss 1.0 and cuboid loss 1.0
- `superdec_cleivr3d_000009` : chamfer loss 10.0 and cuboid loss 1.0

- (Note: both experiments have been stopped quite early)

25.2. Visualize experiments outputs

Experiment 008 (after 52 epochs):

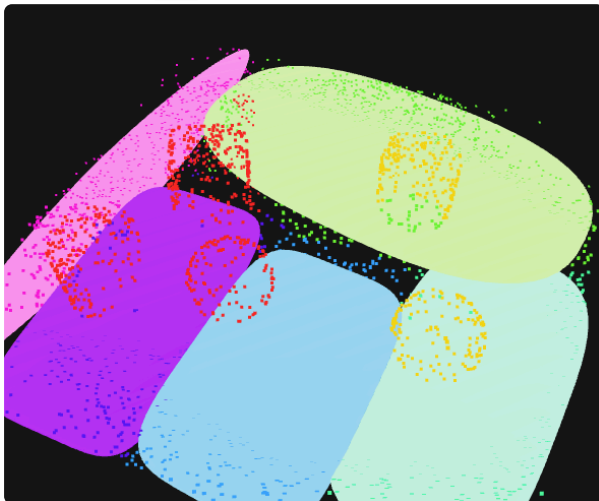


--> first decent results on CLEVR-3D, only when there are few objects in the scene, the ground plane is often covered by multiple superquadrics

--> all objects are covered by cubes, which is not very accurate yet

--> but also training was pretty short yet, so this will probably improve a lot

Experiment 009 (after 54 epochs):



--> higher Chamfer loss overpowers the cuboid loss and results are again similar to the previous results that I had without a cuboid loss

25.2. Fix bug in cuboid loss and start another experiment

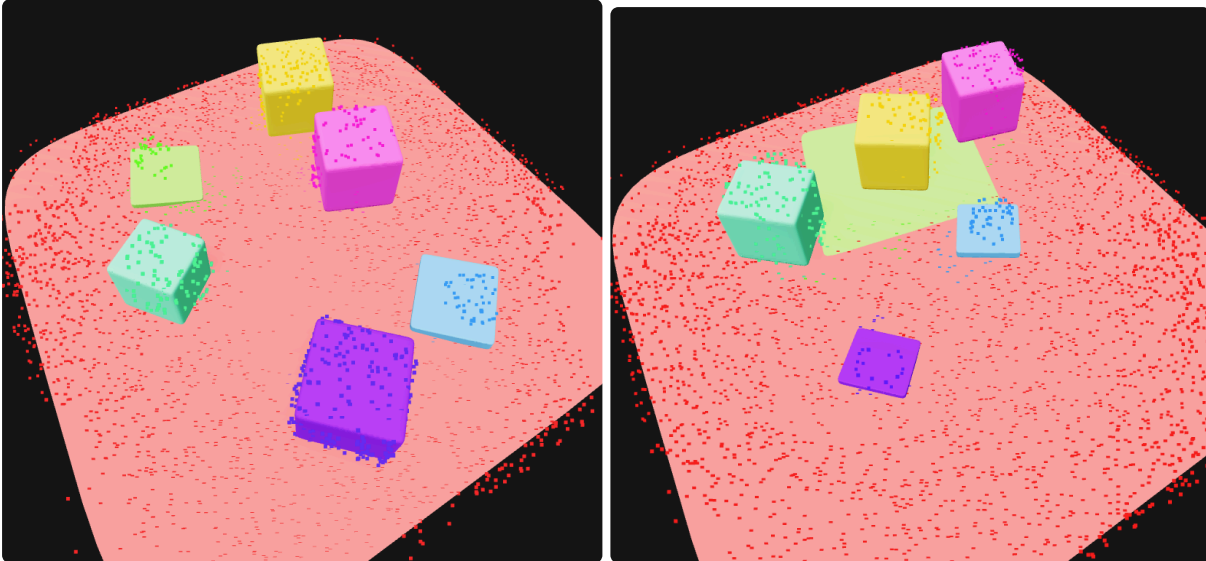
The bug was that in the coordinate transformation of the normals, the superquadric translation was added. In the loss, the normals are anyways only used in a cosine similarity (where they are re-normalized). The bug effect hence mostly depends on the translation magnitude relative to the unit scale of the normals before the coordinate transform.

As in my experiments on SuperDec on CLEVR-3D, most SuperQuadrics are anyways more or less in the center and therefore have rather small translations, the bug is probably not too crucial.

superdec_clevr3d_000010 :

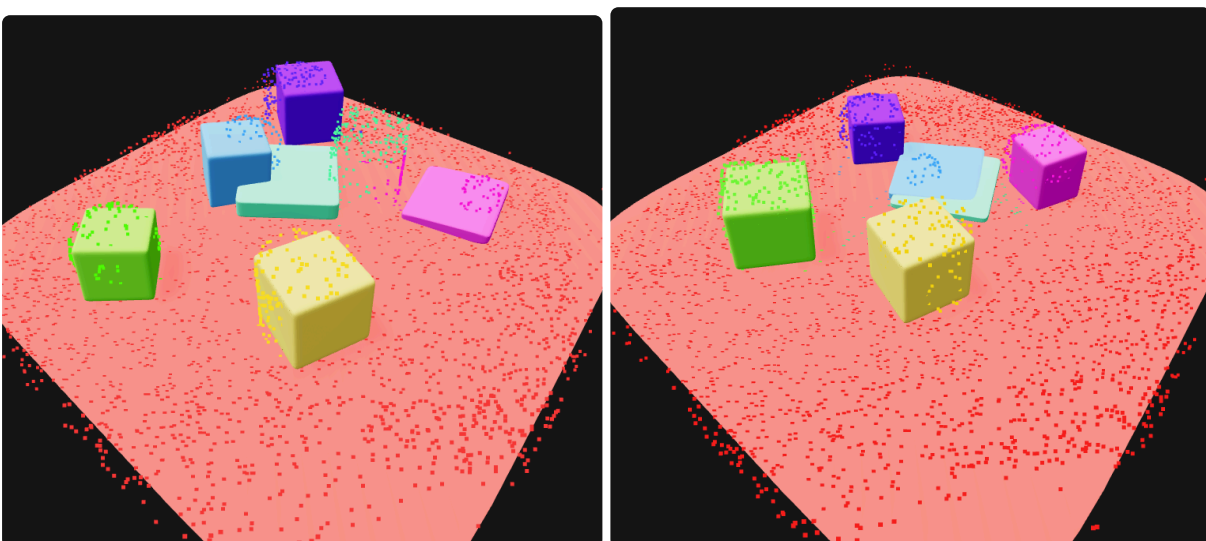
- chamfer loss 1.0 and cuboid loss 1.0 and fix bug in normals coordinate system transformation
- Baseline: 008
- 0010 was trained much longer than 008
- hence: 0010 can be seen as my first real functional SuperDec-CLEVR-3D-model and is a good baseline for later

Visualization of reconstructions after 150 epochs:



--> Reconstructions look good, but when there are too few objects, than unnecessary superquadrics are placed at ground plane, and objects still don't have accurate shapes

Visualization of reconstructions after 281 epochs:



--> looks okay but still not great..

Logbook

Date	Log	Documentation	Source
03.03.2026	Visualize, analyze and document experiments; conclusion: SuperObjectSplat-Quad works okayish but needs more tuning of loss weights depending on scene scale	Link This note Link	Link
25.02.2026	Fix coord transform bug in SuperDec cuboid loss, start another experiment	This note	Link
25.02.2026	Visualize first experiments of SuperDec on CLEVR-3D with cuboid loss	This note	Link
24.02.2026	Compute normals in CLEVR-3D dataloader and start experiments on CLEVR-3D with cuboid loss	This note	Link

Other Mentions

Note	Category	Note created
Playing with SuperDec	Projects	28.11.2025